

Lab 2 : Docker Container Lifecycle and Management Using Nginx

Step 1: Check Docker Installation

docker --version

Output:-

```
C:\>docker --version
Docker version 28.5.1, build e180ab8
```

Step 2: Search for Nginx

docker search nginx

Output:-

```
C:\>docker search nginx
```

NAME	DESCRIPTION	STARS	OFFICIAL
nginx	Official build of Nginx.	21337	[OK]
nginx/nginx-ingress	NGINX and NGINX Plus Ingress Controllers fo...	121	
nginx/nginx-prometheus-exporter	NGINX Prometheus Exporter for NGINX and NGIN...	52	
nginx/nginx-ingress-operator	NGINX Ingress Operator for NGINX and NGINX P...	4	
nginx/nginxaas-loadbalancer-kubernetes		1	
bitnamicharts/nginx	Bitnami Helm chart for NGINX Open Source	4	
ubuntu/nginx	Nginx, a high-performance reverse proxy & we...	141	
kasmweb/nginx	An Nginx image based off nginx:alpine and in...	9	
rancher/nginx		4	
linuxserver/nginx	An Nginx container, brought to you by LinuxS...	236	
dtagevsec/nginx	T-Pot Nginx	0	
paketobuildpacks/nginx		0	
vmware/nginx		3	
gluufederation/nginx	A customized NGINX image containing a consu...	1	
cleanstart/nginx	Secure by Design, Built for Speed, Hardened ...	0	
antrea/nginx	Nginx server used for Antrea e2e testing	0	
activestate/nginx	ActiveState's customizable, low-to-no vulner...	0	
intel/nginx		0	
docksal/nginx	Nginx service image for Docksal	1	
geokrety/nginx	Our customized nginx image	0	
circleci/nginx	This image is for internal use	2	
ilios/nginx	Nginx customized to run Ilios along with the...	0	
corpusops/nginx	https://github.com/corpusops/docker-images/	1	
wayofdev/nginx	Nginx Docker image for PHP development. SSL-...	0	
dockette/nginx	Nginx SSL / HSTS / HTTP2	3	

Step 3: Download Nginx

docker pull nginx

Output:-

```
C:\>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
45ef33eb91c8: Pull complete
062e450697fa: Pull complete
8d2092eabbc5: Pull complete
3b9f3509a826: Pull complete
6a24a6ebcf30: Pull complete
0b19be717994: Pull complete
39694cde734e: Pull complete
Digest: sha256:c90e4ea5905ee69515d84cd5520e11d204df06cce71a312666cebbbc4b9b68267
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

Step 4: View Images

docker images

Output:-

```
C:\>docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
nginx	latest	c90e4ea5905e	8 hours ago	238MB
calculatorapp	v1	0c9e8d340228	10 days ago	203MB
simpleapp	v1	08cb0fff836b	10 days ago	203MB
gupta1725/flask-demo	v1	a3d9b47db284	4 months ago	1.6GB
age-calculator	latest	f215ad5960d9	5 months ago	212MB
jenkins/jenkins	lts	1e407df1a9bc	7 months ago	822MB
student-login-app	latest	d615561ce096	7 months ago	1.6GB
<none>	<none>	7b1c37827827	8 months ago	809MB
alpine	latest	4b7ce07002c6	9 months ago	12.8MB
hello-world	latest	56433a6be3fd	11 months ago	20.3kB
jenkins/jenkins	jdk21	69da3bdc6927	11 months ago	798MB

Step 5: Run Nginx

docker run nginx

Output:-

```
C:\>docker run nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/14 09:20:09 [notice] 1#1: using the "epoll" event method
2026/07/14 09:20:09 [notice] 1#1: nginx/1.31.2
2026/07/14 09:20:09 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/14 09:20:09 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/07/14 09:20:09 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/14 09:20:09 [notice] 1#1: start worker processes
2026/07/14 09:20:09 [notice] 1#1: start worker process 29
2026/07/14 09:20:09 [notice] 1#1: start worker process 30
2026/07/14 09:20:09 [notice] 1#1: start worker process 31
2026/07/14 09:20:09 [notice] 1#1: start worker process 32
2026/07/14 09:20:09 [notice] 1#1: start worker process 33
2026/07/14 09:20:09 [notice] 1#1: start worker process 34
2026/07/14 09:20:09 [notice] 1#1: start worker process 35
2026/07/14 09:20:09 [notice] 1#1: start worker process 36
2026/07/14 09:20:09 [notice] 1#1: start worker process 37
2026/07/14 09:20:09 [notice] 1#1: start worker process 38
2026/07/14 09:20:09 [notice] 1#1: start worker process 39
2026/07/14 09:20:09 [notice] 1#1: start worker process 40
|
```

Step A

docker.desktopPERSONAL

Q Search

Ctrl+K

2

Sign in

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP ToolkitBETA

Docker Hub

Docker Scout

Extensions

Images

LocalMy Hub

4.29 GB / 6.79 GB in use11 images

Last refresh: 10 days ago

Q Search

	Name	Tag	Image ID	Created	Size	Actions
☐	calculatorapp	v1	0c9e8d340228	10 days ago	202.68 MB	<a> <a> <a>
☐	simpleapp	v1	08cb0fff836b	10 days ago	202.67 MB	<a> <a> <a>
☐	gupta1725/flask-demo	v1	a3d9b47db284	5 months ago	1.59 GB	<a> <a> <a>
☐	age-calculator	latest	f215ad5960d9	5 months ago	212.28 MB	<a> <a> <a>
☐	jenkins/jenkins	lts	1e407df1a9bc	7 months ago	821.5 MB	<a> <a> <a>
☐	student-login-app	latest	d615561ce096	8 months ago	1.6 GB	<a> <a> <a>
☐	<none>	<none>	7b1c37827827	8 months ago	809.24 MB	<a> <a> <a>
☐	alpine	latest	4b7ce07002c6	9 months ago	12.79 MB	<a> <a> <a>
☐	hello-world	latest	56433a6be3fd	11 months ago	20.34 KB	<a> <a> <a>
☐	jenkins/jenkins	jdk21	69da3bdc6927	1 year ago	797.56 MB	<a> <a> <a>
☐	nginx	latest	c90e4ea5905e	8 hours ago	237.69 MB	<a> <a> <a>

Showing 11 items

Engine running

RAM 0.99 GB CPU 3.25% Disk: 10.07 GB used (limit 1006.85 GB)

TerminalUpdate available

Step B

docker.desktopPERSONAL

Q Search

Ctrl+K

2

Sign in

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP ToolkitBETA

Docker Hub

Docker Scout

Extensions

Containers

Container CPU usage0.00% / 1200% (12 CPUs available)

Container memory usage10.55MB / 7.43GB

Show charts

Q Search

Only show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	quirky_panini	93f4d511f3ac	gupta1725/flask-demo	5000:5000	0%	4 months ago	<a> <a> <a>
☐	focused_mclean	2eb1c9642e03	age-calculator	5000:5000	0%	5 months ago	<a> <a> <a>
☐	eager_rubin	c410a88655cb	jenkins/jenkins:lts		0%	6 months ago	<a> <a> <a>
☐	vol-demo	9e8215bc4a0a	alpine		0%	7 months ago	<a> <a> <a>
☐	heuristic_turing	ce65ad33296c	student-login-app	5000:5000	0%	8 months ago	<a> <a> <a>
☐	recurring_pasteur	1888385ae77a	hello-world		0%	8 months ago	<a> <a> <a>
☐	jenkins	26a7ac0e3cd4	jenkins/jenkins:lts	50000:50000	0%	6 months ago	<a> <a> <a>
☐	adoring_feynman	1de25c9dc4b8	student-login-app	5000:5000	0%	8 months ago	<a> <a> <a>
☐	flamboyant_tesla	6a3ba2e46826	hello-world		0%	8 months ago	<a> <a> <a>
☐	mjenkins	509c77131e1e	jenkins/jenkins:jdk21	50000:50000	0%	1 year ago	<a> <a> <a>
☐	naughty_cartwright	53982f76b746	nginx		0%	14 minutes ago	<a> <a> <a>
☐	magical_germain	db90dee0a026	nginx:latest		0%	7 seconds ago	<a> <a> <a>

Showing 18 items

Engine running

RAM 0.99 GB CPU 0.00% Disk: 10.07 GB used (limit 1006.85 GB)

TerminalUpdate available

Step C

The screenshot shows the Docker Desktop application window. The left sidebar contains navigation options: Containers, Images, Volumes, Kubernetes, Builds, Models, MCP Toolkit (BETA), Docker Hub, Docker Scout, and Extensions. The main area displays the 'magical_germain' container, which is running. The 'Logs' tab is selected, showing a series of messages from the container's entrypoint script. The messages include configuration steps for nginx, such as enabling IPv6 listening and sourcing environment files. A status bar at the bottom indicates the engine is running with 0.90 GB RAM and 0.08% CPU usage.

docker.desktop PERSONAL

Search

Containers / magical_germain

magical_germain

db90dee0a026 nginx:latest

STATUS

Running (1 minute ago)

Logs Inspect Bind mounts Exec Files Stats

/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/14 09:34:14 [notice] 1#1: using the "epoll" event method
2026/07/14 09:34:14 [notice] 1#1: nginx/1.31.2
2026/07/14 09:34:14 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/14 09:34:14 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/07/14 09:34:14 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/14 09:34:14 [notice] 1#1: start worker processes
2026/07/14 09:34:14 [notice] 1#1: start worker process 29
2026/07/14 09:34:14 [notice] 1#1: start worker process 30
2026/07/14 09:34:14 [notice] 1#1: start worker process 31
2026/07/14 09:34:14 [notice] 1#1: start worker process 32
2026/07/14 09:34:14 [notice] 1#1: start worker process 33
2026/07/14 09:34:14 [notice] 1#1: start worker process 34
2026/07/14 09:34:14 [notice] 1#1: start worker process 35
2026/07/14 09:34:14 [notice] 1#1: start worker process 36
2026/07/14 09:34:14 [notice] 1#1: start worker process 37
2026/07/14 09:34:14 [notice] 1#1: start worker process 38
2026/07/14 09:34:14 [notice] 1#1: start worker process 39
2026/07/14 09:34:14 [notice] 1#1: start worker process 40

Engine running

RAM 0.90 GB CPU 0.08% Disk: 10.07 GB used (limit 1006.85 GB)

_ Terminal Update available

Step D

```
2026/07/14 09:36:29 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/07/14 09:36:29 [notice] 30#30: exiting
2026/07/14 09:36:29 [notice] 34#34: exiting
2026/07/14 09:36:29 [notice] 31#31: exiting
2026/07/14 09:36:29 [notice] 30#30: exit
2026/07/14 09:36:29 [notice] 29#29: exiting
2026/07/14 09:36:29 [notice] 33#33: exiting
2026/07/14 09:36:29 [notice] 37#37: exiting
2026/07/14 09:36:29 [notice] 39#39: exiting
2026/07/14 09:36:29 [notice] 32#32: exiting
2026/07/14 09:36:29 [notice] 31#31: exit
2026/07/14 09:36:29 [notice] 29#29: exit
2026/07/14 09:36:29 [notice] 36#36: exiting
2026/07/14 09:36:29 [notice] 34#34: exit
2026/07/14 09:36:29 [notice] 33#33: exit
2026/07/14 09:36:29 [notice] 39#39: exit
2026/07/14 09:36:29 [notice] 27#37: exit
2026/07/14 09:36:29 [notice] 36#36: exit
2026/07/14 09:36:29 [notice] 38#38: exiting
2026/07/14 09:36:29 [notice] 32#32: exit
2026/07/14 09:36:29 [notice] 38#38: exit
2026/07/14 09:36:29 [notice] 40#40: exiting
2026/07/14 09:36:29 [notice] 40#40: exit
2026/07/14 09:36:29 [notice] 35#35: exiting
2026/07/14 09:36:29 [notice] 35#35: exit
2026/07/14 09:36:29 [notice] 1#1: signal 14 (SIGALRM) received
2026/07/14 09:36:29 [notice] 1#1: signal 17 (SIGCHLD) received from 29
2026/07/14 09:36:29 [notice] 1#1: worker process 29 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: signal 29 (SIGIO) received
2026/07/14 09:36:29 [notice] 1#1: signal 17 (SIGCHLD) received from 40
2026/07/14 09:36:29 [notice] 1#1: worker process 40 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: signal 29 (SIGIO) received
2026/07/14 09:36:29 [notice] 1#1: signal 17 (SIGCHLD) received from 34
2026/07/14 09:36:29 [notice] 1#1: worker process 30 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: worker process 31 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: worker process 32 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: worker process 34 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: worker process 37 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: signal 29 (SIGIO) received
2026/07/14 09:36:29 [notice] 1#1: signal 17 (SIGCHLD) received from 32
2026/07/14 09:36:29 [notice] 1#1: worker process 33 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: worker process 36 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: worker process 39 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: signal 29 (SIGIO) received
2026/07/14 09:36:29 [notice] 1#1: signal 17 (SIGCHLD) received from 38
2026/07/14 09:36:29 [notice] 1#1: worker process 38 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: signal 29 (SIGIO) received
2026/07/14 09:36:29 [notice] 1#1: signal 17 (SIGCHLD) received from 35
2026/07/14 09:36:29 [notice] 1#1: worker process 35 exited with code 0
2026/07/14 09:36:29 [notice] 1#1: exit
```

Step 6: Detached Mode

`docker run -d nginx`

Output:-

```
C:\>docker run -d nginx
5c9a1e9dd652dfe12d2c24154a3eb43bcc09c30d7acf15e26930d53e6d488cc0
```

Step 7: Name the Container

`docker run -d --name mynginx nginx`

Output:-

```
C:\>docker run -d --name mynginx nginx
04d538b8dbb4fd58251fc5dffa2530087c97241d3a1443830ff543526fe665627
```

Step 8: Publish a Port

`docker run -d --name webserver -p 8080:80 nginx`

Output:-

```
C:\>docker run -d --name webserver -p 8080:80 nginx
234b1b1e96cc0a41ddf68c8f7b8f83eeb2c05357c6950943f24a4cef9a1b9255
```

Step 9: List Running Containers

`docker ps`

Output:-

```
C:\>docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
234b1b1e96cc	nginx	"/docker-entrypoint..."	3 minutes ago	Up 3 minutes	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	webserver
04d538b8dbb4	nginx	"/docker-entrypoint..."	5 minutes ago	Up 5 minutes	80/tcp	mynginx
5c9a1e9dd652	nginx	"/docker-entrypoint..."	7 minutes ago	Up 7 minutes	80/tcp	brave_heisenberg
db90dee0a026	nginx:latest	"/docker-entrypoint..."	13 minutes ago	Up 13 minutes	80/tcp	magical_germain

Step 10: Stop Container

docker stop mynginx

Output:-

```
C:\>docker stop mynginx
mynginx
```

Step 11: Start Container Again

docker start mynginx

Output:-

```
C:\>docker start mynginx
mynginx
```

Step 12: Restart

docker restart mynginx

Output:-

```
C:\>docker restart mynginx
mynginx
```

Step 13: View Logs

`docker logs mynginx`

Output:-

```
C:\>docker logs mynginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/14 09:41:56 [notice] 1#1: using the "epoll" event method
2026/07/14 09:41:56 [notice] 1#1: nginx/1.31.2
2026/07/14 09:41:56 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/14 09:41:56 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/07/14 09:41:56 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/14 09:41:56 [notice] 1#1: start worker processes
2026/07/14 09:41:56 [notice] 1#1: start worker process 29
2026/07/14 09:41:56 [notice] 1#1: start worker process 30
2026/07/14 09:41:56 [notice] 1#1: start worker process 31
2026/07/14 09:41:56 [notice] 1#1: start worker process 32
2026/07/14 09:41:56 [notice] 1#1: start worker process 33
2026/07/14 09:41:56 [notice] 1#1: start worker process 34
2026/07/14 09:41:56 [notice] 1#1: start worker process 35
2026/07/14 09:41:56 [notice] 1#1: start worker process 36
2026/07/14 09:41:56 [notice] 1#1: start worker process 37
2026/07/14 09:41:56 [notice] 1#1: start worker process 38
2026/07/14 09:41:56 [notice] 1#1: start worker process 39
```

Step 14: Enter the Container

`docker exec -it mynginx bash`

Output:-

```
C:\>docker exec -it mynginx bash
root@04d538b8dbb4:/# pwd
/
root@04d538b8dbb4:/# ls
bin  dev  docker-entrypoint.sh  home  lib64  mnt  proc  run  srv  tmp  var
boot  docker-entrypoint.d  etc  lib  media  opt  root  sbin  sys  usr
root@04d538b8dbb4:/# |
```


Step 15: Remove the Container

`docker rm mynginx`

Output:-

```
C:\>docker rm mynginx
Error response from daemon: cannot remove container "mynginx": container is running: stop the container before removing or force remove
```

Step 16: Remove the Image

`docker rmi nginx`

Output:-

```
C:\>docker rmi nginx
Error response from daemon: conflict: unable to delete nginx:latest (must be forced) - container 04d538b8dbb4 is using its referenced image c90e4ea5905e
```